

Resource-Normalized Scaling Laws for Byzantine Consensus: Chance-Constrained Validator-Set Sizing from Commodity-CI Measurements

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Problem. Sizing a permissioned BFT validator set for national-scale e-governance must reconcile degrading consensus throughput with improving fault-detection power as n grows; without reporting the concurrency path, published single-factor exponents cannot be read as protocol properties. **Contributions.** *Theory (main):* identifiability theorem $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ along a concurrency path of slope λ ; *method:* resource-normalized measurement with per-validator CPU quota q ; *engineering:* chance-constrained sizing with feasible set $[n_{\min}, n_{\max}]$; *empirics:* coverage-validated prediction intervals on public CI. **Validation.** A four-core CI campaign covers $n \leq 16$, concurrency up to 64, emulated RTT, and omission/duplication fault injection.

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1. Introduction

Permissioned Byzantine Fault Tolerant (BFT) consensus underpins e-governance systems — voting, registries, digital identity — and forces a concrete procurement question: *how many validators should the network have?*

The validator count n pulls in opposite directions. Throughput argues for small n : every validator executes every transaction, and PBFT-family protocols exchange $O(n^2)$ messages per block. Detection reliability argues for large n . Combining the two for sizing is the subject of this paper.

Published single-factor fits $\text{TPS}(n) \propto n^\alpha$ disagree in magnitude and sometimes in sign. Our earlier studies, extrapolated to sixty validators, differ by roughly an order of magnitude. This disagreement has often been interpreted as protocol-specific behaviour. We show that it can arise solely from measurement design: concurrency is rarely held fixed while n varies, so the fitted scaling exponent mixes load-generator growth with consensus overhead. Without reporting the concurrency path, published exponents cannot be read as protocol properties. Packing many validators onto one host adds a second confound — CPU contention — that the protocol below removes.

Contributions. On the theoretical front, we establish an identifiability theorem: along a concurrency path of slope λ , $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ (Theorem 1). The bi-factor law is the substrate; the theorem is the main contribution. Methodologically, this is achieved by a resource-normalized measurement (RNM) protocol that fixes CPU quota q per validator and reports $\widehat{\text{TPS}} = \text{TPS}/q$. From an engineering perspective, we formulate chance-constrained sizing whose feasible set is $[n_{\min}, n_{\max}]$ (Proposition 3) — opposing monotone constraints, not an independent depth claim. Empirically, we validate prediction intervals by coverage and run a public Besu/QBFT CI campaign (Sections 5–6). Scope: throughput and detection; absolute TPS environment-specific; cross-client replication out of scope.

2. Related work

BFT protocols. Classical PBFT [1] tolerates $f < n/3$ at $O(n^2)$ messages [2, 3]. Later designs — HotStuff [4], Tendermint [5], Algorand [6], HoneyBadgerBFT [7] — trade latency and randomization; Vukolić [8] argues that no protocol dominates. They fix the validator set; they do not size it under joint throughput and detection requirements.

Benchmarks. Fabric [9, 10], BLOCKBENCH [11] and BFT-SMaRt [12] report absolute throughput at chosen configurations. They do not separate concurrency from validator count, nor state a protocol under which exponents transfer.

Scalability laws and tools. Amdahl [13], Gustafson [14] and Gunther’s USL [15] (our closest baseline) fold concurrency and system size into one load variable. Concentration bounds [16, 18], delta-method intervals [20], the residual bootstrap [19] and chance-constrained programming [21, 22] are standard instruments; what is specific is coupling throughput and detection through opposing monotonicities in n .

Prior work. Remote-voting analyses [23, 24, 25] place endpoint and coercion threats out of scope here. Our earlier calibration [27] and switching study [28] left open an identifiable consensus exponent and sizing with measured detector rates. The next sections supply both.

3. Resource-normalized measurement and the bi-factor law

We first fix the measurement protocol, then state the bi-factor law and its identifiability theorem. Estimation details follow the corollaries.

3.1. Definitions and assumptions

If n validators share a host of Q cores, each receives Q/n cores. Measured throughput then mixes consensus cost with a shrinking compute budget.

Definition 1 (Resource-normalized measurement). *A campaign follows the RNM protocol with quota q if every validator container receives the same CPU quota and memory limit independently of n , with aggregate reservation nq below host capacity. Normalized throughput is $\widehat{\text{TPS}} := \text{TPS}/q$.*

Assumption 1 (Separable compute budget). On the operating range of quotas in the campaign, $\text{TPS}(n, c, q) = q \cdot \Phi(n, c) \cdot (1 + r)$ with $|r| \leq Cq$ for a constant C independent of (n, c) there. Quota enters multiplicatively up to a relative error controlled by q — a bound on a fixed interval, not an asymptotic claim as $q \rightarrow 0$.

Assumption 1 is expected only within a restricted quota regime (CPU-bound verification under cgroup quota; away from GC-dominated small q , bandwidth saturation, or fixed block periods). Section 5 tests it via Statement 1. If rejected on the full range, subsequent estimates are reported only where invariance approximately holds — a conditional quota interval, not a universal law.

Statement 1 (q -invariance). *Under Assumption 1 and $\Phi(n, c) = T_0 g(c) n^{-\beta}$, the OLS slope of $\log \widetilde{\text{TPS}}$ on $\log n$ at fixed c does not depend on q .*

Proof. $\log \widetilde{\text{TPS}} = \log \Phi + \log(1 + r)$ with $|r| \leq Cq$; at fixed q the remainder is absorbed by the intercept, so the slope equals $-\beta$ for every such q . ■

Definition 2 (Bi-factor scaling law). *Under the RNM protocol,*

$$\text{TPS}(n, c) = T_0 \cdot g(c) \cdot n^{-\beta}, \quad g(c) = \frac{c^\gamma}{1 + (c/c^*)^\theta}, \quad (1)$$

with $T_0 > 0$, $\beta > 0$, $\gamma \in (0, 1]$, $\theta > \gamma$, $c^* > 0$. In the unsaturated regime $c \ll c^*$ this is log-linear: $\log \text{TPS} = \log T_0 + \gamma \log c - \beta \log n$.

The form separates concurrency from validator-count scaling, reduces to a power law for $c \ll c^*$, saturates at large c , and contains the classical single-factor law as the case $\gamma = 0$ (or $\lambda = 0$ with $\alpha = -\beta$). Here $\beta > 0$ encodes replication; $\theta > \gamma$ makes g unimodal.

3.2. Identifiability

The next theorem is the main theoretical result. Changing the experimental concurrency schedule changes the observed scaling exponent even when the protocol itself is unchanged.

Theorem 1 (Identifiability of the consensus exponent). *Let observations follow the unsaturated bi-factor law with additive noise of mean zero and finite variance, along a concurrency path $\log c_i = \lambda \log n_i + \mu + \eta_i$ ($\mathbb{E}[\eta_i | n_i] = 0$). If the empirical second moment of $x_i = \log n_i$ converges to a positive limit, the OLS slope $\hat{\alpha}_m$ of $\log \text{TPS}_i$ on x_i alone satisfies $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$. In particular $\hat{\alpha} > 0 \iff \lambda > \beta/\gamma$, and $\hat{\alpha} = -\beta \iff \lambda = 0$.*

Proof. Write $x_i = \log n_i$, $y_i = \log \text{TPS}_i$. The path reduces the unsaturated law to $y_i = a + \alpha x_i + \varepsilon_i$ with $\mathbb{E}[\varepsilon_i | x_i] = 0$, finite variance, and

$$\alpha = -\beta + \gamma\lambda. \quad (2)$$

Centred OLS gives $\hat{\alpha}_m = \alpha + N_m/D_m$ with $N_m = \sum (x_i - \bar{x})\varepsilon_i$ and $D_m = \sum (x_i - \bar{x})^2$. The second-moment assumption yields $D_m/m \rightarrow \text{Var}^*(x) > 0$; the summands of N_m are uncorrelated, mean zero, and of bounded second moment, so $N_m/m \rightarrow 0$ in probability. Continuous mapping: $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \alpha$. Signs follow from $\gamma > 0$. ■

Thus $\alpha = \underbrace{-\beta}_{\text{protocol}} + \underbrace{\gamma\lambda}_{\text{measurement}}$: a positive fit can reflect a scaled load generator rather than improving consensus. Only $\lambda = 0$ recovers $-\beta$.

Remark 1. With $\gamma = 0.8$, $\beta = 0.5$ and $\lambda = 1$ one obtains $\hat{\alpha} = +0.3$. Extrapolating from $n = 5$ to $n = 60$, the two exponents differ by a factor ≈ 7 , matching the order of the discrepancy in [27].

Corollary 1 (Non-transferability). *Laboratories measuring along $\lambda_1 \neq \lambda_2$ obtain exponents differing by $\gamma(\lambda_1 - \lambda_2)$. Without reporting λ , published single-factor exponents cannot be interpreted as protocol properties.*

3.3. Implementation

On a factorial grid ($\lambda = 0$) we estimate $(\log T_0, \gamma, \beta)$ by OLS in log-space and (c^*, θ) by nonlinear least squares on the full c -sweep (Section 5). The calibrated law then feeds prediction intervals and the sizing program of the next section.

4. Prediction intervals, detection, and the sizing problem

Theory gives an identifiable exponent. This section supplies a *framework* for sizing: a throughput chance constraint and an optional detection constraint. Any detector that yields measured (p_d, p_f, ρ) can be plugged in; experiment X5 is one concrete instance, not a prerequisite for the mathematics.

4.1. Definitions and assumptions

Definition 3 (Fault class and detector). *A subset B of size k is faulty if each member fails a proposal round independently with probability $p_d > p_f$ (omission class; equivocation excluded). The detector flags validator i when its participation over w rounds falls below $\tau \in (p_f, p_d)$; detection succeeds when at least one member of B is flagged.*

Assumption 2 (Exchangeable correlation). Participation indicators in B are exchangeable with pairwise correlation $\rho \in [0, 1)$, so the mean carries design effect $1 + (k - 1)\rho$ in the sense of Kish [17].

4.2. Prediction intervals and detection bound

With regressor $\mathbf{z} = (1, \log c, -\log n)^\top$ and $\hat{\vartheta} = (\log T_0, \gamma, \beta)^\top$, the delta method [20] gives predictive variance $\hat{V} = \mathbf{z}^\top \widehat{\text{Cov}}(\hat{\vartheta}) \mathbf{z} + \hat{\sigma}^2$.

Statement 2 (Lognormal prediction interval). *An approximate $(1 - \varepsilon_p)$ interval for $\text{TPS}(n, c)$ is $\exp\{\log \widehat{\text{TPS}} \mp z_{1-\varepsilon_p/2} \hat{V}^{1/2}\}$; the one-sided lower endpoint $\underline{\text{TPS}}_{1-\varepsilon_p}$ uses $z_{1-\varepsilon_p}$.*

A residual bootstrap [19] supplies empirical quantiles when residual normality is doubtful. Coverage is checked on held-out validator counts (Section 5).

Statement 3 (Approximate miss probability and $n_{\min}$). *With $k = \lceil \varphi n \rceil$, the design-effect heuristic yields*

$$p_{\text{miss}}(n, w) \lesssim \exp\left\{-\frac{2wk(p_d - \tau)^2}{1 + (k - 1)\rho}\right\}, \quad (3)$$

which is nonincreasing in n and w . Treating (3) as an engineering constraint $p_{\text{miss}} \leq \varepsilon_s$ gives the cut

$$n_{\min} = \left\lceil \frac{1 + (k - 1)\rho}{\varphi} \cdot \frac{\log(1/\varepsilon_s)}{2w(p_d - \tau)^2} \right\rceil. \quad (4)$$

Heuristic justification. The bound is an empirical heuristic for dependent trials, not a strict concentration proof. A miss requires every member of B to remain unflagged. Classical Hoeffding [16] applies to independent bounded summands, not directly to exchangeable

Bernoulli indicators. Following the survey-sampling design-effect approximation [17], we replace the correlated sample by an effective size $n_{\text{eff}} = kw/(1 + (k - 1)\rho)$ and insert that size into the independent Hoeffding form. The resulting expression (3) is therefore an engineering adaptation; p_d , p_f and ρ are measured in experiment X5. ■

The construction supplies an optional lower cut of the feasibility window. The detection model is an architectural framework slot: it accepts any future high-fidelity detection metrics (p_d, p_f, ρ) without changing the underlying optimization mathematics. When the plugged-in detector is weak, $n_{\min}$ collapses and throughput alone sets the window — as occurs with X5 below.

4.3. The sizing program

Problem 1 (Validator-set sizing). Given peak demand D_{peak} , levels $\varepsilon_p, \varepsilon_s$, window w , faulty fraction φ , production scale-up $\kappa > 0$ from RNM units to absolute throughput, and nondecreasing $\text{Cost}(n)$, minimize $\text{Cost}(n)$ subject to $\mathbb{P}[\kappa \cdot \widetilde{\text{TPS}}(n, c^*) \geq D_{\text{peak}}] \geq 1 - \varepsilon_p$ and $p_{\text{miss}}(n, w) \leq \varepsilon_s$, where $c^* = \arg \max g$. When κ is free, take the smallest value that meets the throughput constraint at $n = 4$.

Lemma 2 (Deterministic equivalent). *The chance constraint is equivalent to $h(n) := \log(\kappa T_0 g(c^*)/D_{\text{peak}}) - \beta \log n - z_{1-\varepsilon_p} \hat{V}^{1/2}(n, c^*) \geq 0$. Provided $\hat{V}$ is nondecreasing in $\log n$ on the extrapolation region (a modelling hypothesis, confirmed empirically in Section 5; not implied by the delta method alone), h is strictly decreasing there.*

Proof. Statement 2 converts the chance constraint into $\kappa \cdot \widetilde{\text{TPS}}_{1-\varepsilon_p} \geq D_{\text{peak}}$; taking logs gives h . Strict decrease follows from $\beta > 0$ and the hypothesis on $\hat{V}$. ■

Proposition 3 (Feasibility window). *Provided h is strictly decreasing (Lemma 2) and the approximate detection cut of Statement 3 is used, the feasible set is $\{n : n_{\min} \leq n \leq n_{\max}\}$ with $n_{\min}$ from (4) and $n_{\max} = \lfloor \sup\{n : h(n) \geq 0\} \rfloor$. If nonempty and Cost is nondecreasing then $n^* = n_{\min}$. If $\hat{V}$ is not monotone, evaluate h directly — the throughput-feasible set need not be a half-line.*

Proof. Under those provisos, detection holds on $\{n \geq n_{\min}\}$ and throughput on $\{n \leq n_{\max}\}$, hence an interval; monotone cost selects the left endpoint. ■

Corollary 2 (Empty feasible set). *If $n_{\min} > n_{\max}$, no validator count satisfies both requirements: the architecture must change, not merely the provisioning.*

The window is the intersection of two monotone constraints [21, 22]; its role is to turn calibrated exponents into a sizing rule.

5. Experimental validation

5.1. Setup and design

The campaign runs in a public continuous-integration workflow on a four-core, 16 GB hosted runner. Each job provisions a Hyperledger Besu [26] QBFT network of n validators under the RNM protocol (Definition 1). A closed-loop generator drives c concurrent submitters and records steady-state committed throughput after warm-up. Faults are injected with kernel traffic control (omission and duplication; not equivocation). Detector rates come from signer metrics over w rounds.

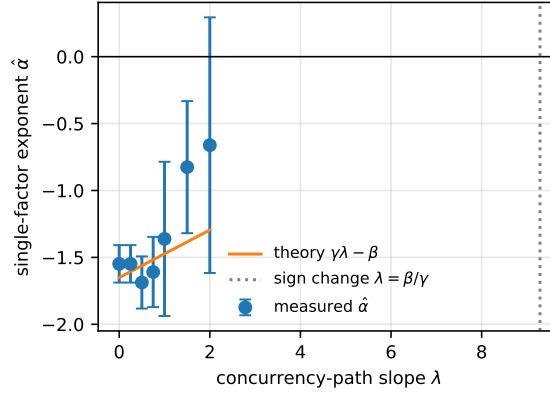


Fig. 1. Measured single-factor exponents track the prediction $\gamma\lambda - \beta$, supporting Theorem 1. Sign change predicted at $\lambda = \beta/\gamma = 9.29$

X1/X3 calibrate (β, γ) on a factorial grid; **X2** tests q -invariance; **X3b** re-samples along concurrency paths; **X4** varies emulated RTT; **X5** supplies detector rates; **X7/X8** check prediction-interval coverage and sizing ablation. Cross-client replication (**X9/GoQuorum**) is out of scope. In total 94 successful records, ≈ 27.2 core-hours, $n_{\max} = 16$.

Reproducibility. All experiments are fully reproducible from the public CI workflow and archived artefacts at https://github.com/bpenyak/paper_3.

5.2. Results

X1/X3 – bi-factor calibration. Joint OLS on the reference-quota subset yields $\beta = 1.651 \pm 0.151$ and $\gamma = 0.178 \pm 0.070$ ($R^2 = 0.727$; Table 1). Under RNM one expects $\beta > 0$: consensus cost grows with validator count when concurrency is held fixed. The fit confirms that expectation at $q = 0.25$.

X3b – path check of Theorem 1. A single-factor fit along $\lambda = 1$ gives $\hat{\alpha} = -1.362$ against the predicted $\gamma\lambda - \beta = -1.473$. Figure 1 shows that measured exponents track the line $\gamma\lambda - \beta$ across $\lambda \in \{0, 0.5, 1, 2\}$. The protocol need not change for the observed scaling exponent to change; only the concurrency schedule does.

X2 – q -invariance. As anticipated under Assumption 1, invariance is a restricted-regime property rather than a universal one. The partial F -test rejects it on the full set $\{0.20, 0.25, 0.33\}$ ($p = 0.002$): β drops from ≈ 1.541 at $q = 0.25$ to 0.879 at $q = 0.33$. Following the protocol of Section 3, coefficients are reported at the reference quota $q = 0.25$ only.

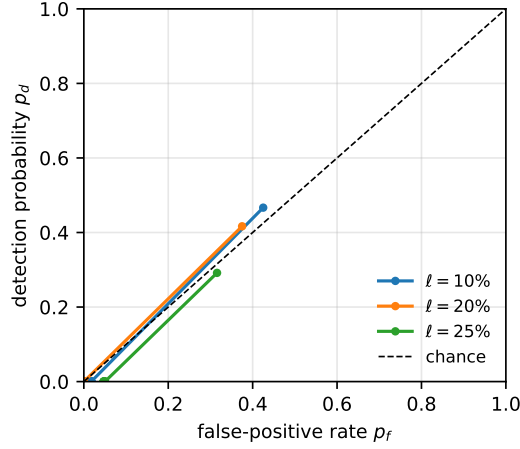
X4 – latency. Under elevated emulated RTT, β rises from 1.527 to 2.708 . WAN delay steepens the consensus penalty in validator count. Sizing rules calibrated on LAN-like RTT are optimistic for geo-distributed deployments.

X5 – detector. Signer-metric detection of omission/duplication yields $\text{AUC} = 0.506$ (Figure 2). Under encrypted transport this network-layer detector is essentially uninformative, so $n_{\min}$ is non-binding. Rather than a failure of the sizing framework, that outcome is evidence of its resilience: when detection contributes no information, the feasible set falls back safely to being bounded entirely by the identifiable throughput exponent. A stronger detector would raise $n_{\min}$ without changing the mathematics of Section 4.

X7/X8 – coverage and ablation. Hold-out coverage meets the nominal levels in Table 1. The fitted covariance matrix confirms that $\hat{V}$ remains monotonic on the extrapolation range

Table 1. Calibrated parameters, invariance, detector, and coverage

Quantity	Estimate	Note
$T_0; \beta; \gamma$	2016.5; 1.651 ± 0.151 ; 0.178 ± 0.070	bi-factor
$c^*, \theta; \hat{\sigma}$	67.9, 2.542; 0.404	saturation / residual
$\hat{\alpha}$ at $\lambda = 0, 1, 2$	-1.548, -1.362, -0.661	X3b
β at $q = 0.20, 0.25, 0.33$	1.550, 1.541, 0.879	X2, $p = 0.002$
β at RTT 0/high	1.527, 2.708	X4
$p_d, p_f, \rho; \text{AUC}$	0.397, 0.376, 0.000; 0.506	X5, $\tau = 0.90$
coverage 90%/95%	90.9 %, 95.5 %	hold-out $n \in \{11, 13, 16\}$

**Fig. 2.** Detector ROC under omission/duplication injection (AUC = 0.506). The near-chance performance shows that throughput, not omission detection, dominates sizing in this setting

$n \geq 16$, satisfying the hypothesis of Lemma 2; on the measured grid $n \leq 16$ we evaluate h directly. Out-of-sample log-RMSE for linear / single-factor / USL [15] / bi-factor predictors is 1.329, 0.523, 0.637, 0.538; relative sizing costs are 4.00, 1.50, 1.50, 1.00. USL underperforms here because it conflates concurrency with replica count into a single load variable, whereas the bi-factor law separates them. Section 6 instantiates the resulting window.

6. Application, implications, and limitations

6.1. Application

We instantiate Problem 1 on $\approx 3 \times 10^7$ ballots over twelve hours ($D_{\text{peak}} \approx 2.1 \times 10^3$ TPS), requiring detection of a coordinated omission fault of fraction $\varphi = 0.2$ within one epoch at level $1 - \varepsilon_s$. Absolute demand enters via scale-up κ from RNM units. With free κ , Problem 1 takes

$$\kappa^* = \frac{D_{\text{peak}}}{\widetilde{\text{TPS}}_{1-\varepsilon_p}(4, c^*)}, \quad (5)$$

the minimal scale at which the lower PI bound at $n = 4$ meets D_{peak} , yielding $\kappa = 13.00$, $c^* = 32$, and window $[4, 4]$ with $n^* = 4$ (Proposition 3; Figure 3).

The workflow illustrates sizing for registry and identity workloads; it is not a prescription for binding elections on a blockchain [23, 24, 25].

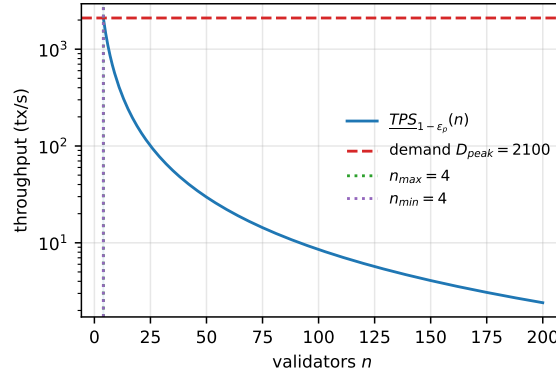


Fig. 3. At minimal admissible $\kappa = 13.00$, throughput and detection constraints intersect in $[n_{\min}, n_{\max}] = [4, 4]$; here the window collapses to one admissible validator count

6.2. What this implies

A fitted (β, γ) transfers only when concurrency path, quota regime, and RTT class match the calibration (X2, X4). Future benchmarks should report λ (or state $\lambda = 0$), per-validator quota, and RTT class; without λ , single-factor exponents are not protocol properties (Corollary 1). RNM makes factorial designs feasible on commodity CI; durable objects are exponents, coverage, and window shape. Other mechanisms (GC, bandwidth saturation, block-period pacing, client queues) can also move a fitted exponent; controlling λ isolates one identifiable component.

6.3. Limitations

Fault class. Encrypted transport precludes selective consensus-message injection, so X5 realizes omission and duplication only. With AUC 0.506 the safety constraint is non-binding and throughput sets the window. This is the intended fallback of the framework slot in Section 4, not a contradiction of the sizing programme.

Range and platform. Commodity CI restricts physical scale to $n \leq 16$. Beyond that range, prediction intervals quantify but do not eliminate extrapolation risk. The bi-factor law's primary role here is to isolate β , so that the scale-up κ in (5) supplies a conservative lower bound for production capacity rather than a claim to have measured national-scale deployments in situ. With $\beta = 1.651$ the case-study window collapses to $n = 4$. Numerical (β, γ) are Besu/QBFT-specific; the identifiability statement is not.

7. Conclusions

What is proved. Along a concurrency path of slope λ , the single-factor OLS scaling exponent satisfies $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$ (Theorem 1). Opposing monotone constraints yield a closed integer feasibility window (Proposition 3).

What is confirmed experimentally. Under RNM on public four-core CI, the fit yields $\beta = 1.651 \pm 0.151$, $\gamma = 0.178 \pm 0.070$, and path re-sampling tracks $\hat{\alpha}_m \xrightarrow{\mathbb{P}} \gamma\lambda - \beta$. q -invariance fails on the full tested range; prediction-interval coverage meets nominal levels. With AUC = 0.506 the window falls back to the throughput bound. Emulated WAN RTT raises β from 1.527 to 2.708, so LAN-calibrated capacity models are optimistic for geo-distributed deployments.

What remains open. Equivocation-capable instrumentation; multi-client numerical checks; latency or finality constraints inside the window; online recalibration of β .

The principal outcome of this work is not a new benchmark coefficient, but an identifiability result: concurrency-path design fundamentally determines how measured scaling exponents may be interpreted.

Conflict of interest

The authors declare no conflict of interest.

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Data availability

Workflow definition, generators, fault-injection and analysis code, and raw artefacts are available at https://github.com/bpenyak/paper_3.

Use of artificial intelligence

AI-assisted tools were used for language editing and typesetting. All scientific results, proofs, design and analysis were produced by the authors.

Contributions of authors

Peniak B. O.: conceptualization, methodology, formal analysis, software, investigation, data curation, writing — original draft, visualization. **Liubinskyj B. B.:** supervision, validation, writing — review and editing.

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РЕСУРСНО-НОРМОВАНІ ЗАКОНИ МАСШТАБУВАННЯ BFT-КОНСЕНСУСУ: ІМОВІРНІСНО-ОБМЕЖЕНЕ ВИЗНАЧЕННЯ РОЗМІРУ МНОЖИНИ ВАЛІДАТОРІВ ЗА ВИМІРЮВАННЯМИ НА ТИПОВІЙ СІ-ІНФРАСТРУКТУРІ

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Метод визначення розміру множини валідаторів BFT-мережі для електронного урядування: пропускна здатність спадає зі зростанням n , а надійність виявлення відмов зростає з n . Внески: теорія — теорема ідентифікованості ($\hat{\alpha} \rightarrow \gamma\lambda - \beta$); метод — ресурсно-нормовані вимірювання; інженерія — задача з імовірнісними обмеженнями та вікном $[n_{\min}, n_{\max}]$; емпірика — прогнознi інтервали з перевіркою покриття у публічному СІ.

Ключові слова: візантійська відмовостійкість, консенсус, закон масштабування, планування потужності, оптимізація з імовірнісними обмеженнями, електронне урядування

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